

PH 507: X-rays as a probe to study material properties

Course Code	PH 507
Course Name	X-rays as a probe to study material properties
L-T-P-C	3-0-0-3
Prerequisites	IC 121 Mechanics of Particles and waves
Intended for	UG /PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **The discovery of x-rays** [1 Lecture]
- **Interaction of radiation with matter** [5 Lectures]
 - Time independent perturbation theory
 - Time dependent perturbation theory
 - Fermi Golden rule
- **State-of-the art Techniques**
 - (a) **X-ray sources**
 - * Conventional laboratory sources
 - * Synchrotron radiation sources
 - (b) **X-ray Diffraction** [10 Lectures + lab visit]
 - * **Theory**
 - Scattering by electrons
 - Scattering by atoms
 - Scattering by unit cell
 - Crystal axes and Reciprocal lattice
 - Structure factors
 - Diffraction Intensity calculations
 - * **Diffraction measurements**
 - Various diffraction geometries
 - This includes geometries used for (a) ambient conditions (b) extreme (high pressure using diamond anvil cell) conditions.
 - Basic idea behind the generation of high pressure conditions in the laboratory.
 - * **Detection systems**
 - * **Applications**- understanding the order-disorder transformation
 - (c) **X-ray absorption Spectroscopy** [7 Lectures]
 - * Basic theory
 - * Experimental Importance of local structural measurements
 - * Local structural link with the physical properties of different materials
 - * X-ray magnetic circular dichroism
 - (d) **Photoemission spectroscopy** [12 Lectures]
 - * Electron Spectroscopy- Basic Concepts
 - * Electron spectrometer Design
 - * Electron spectrum- Qualitative and Quantitative
 - * Different Photoemission spectroscopic techniques
 - Angle integrated photoemission spectroscopy
 - Angle resolved photoemission spectroscopy
 - Spin resolved photoemission spectroscopy
 - Inverse photoemission spectroscopy
 - Application of spectroscopy in Material Science

Text Books

1. Arthur Holly Compton, **X-rays in Theory and Experiment**, Samuel King Allison, 1935.
2. B.D. Cullity, **Elements of x-ray diffraction**, 3rd Edition, Prentice Hall, 2001.
3. D.C. Koningsberger and R.Prins, (Eds.), **X-ray absorption: Principles, Applications, Techniques of EXAFS, SEXAFS and XANES**, 1988.
4. Stephan Hufner, **Photoelectron Spectroscopy**, Springer, 2003.